

BHarath Health Systems

Complete System Architecture & Product Documentation

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9 Portals	1 Backend API	1 Database	∞ Tenants
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Document Type	System Architecture & Product Blueprint
Platform	BHarath Health Systems (BharatCliniq)
Serves	Clinics and Hospitals across India
Tech Stack	FastAPI · PostgreSQL · React (Vite) · Render · Vercel
Status	Active Development — v1.0
Prepared by	Engineering Team

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Chapter 1 — Mission & Vision

1.1 Who We Are

BHarath Health Systems is a multi-tenant SaaS healthcare platform built specifically for the Indian market. We serve both **clinics and hospitals** of all sizes — from a single-doctor clinic in a small town to a multi-branch hospital chain in a major city. Every feature, workflow, and design decision is shaped by the realities of Indian healthcare: cash-dominant billing, GST compliance, multi-language patient populations, variable connectivity, and the need for affordable, high-quality digital tools.

1.2 The Problem We Solve

Indian clinics and hospitals face three critical challenges with existing software:

Too Expensive	GoFrugal, Practo, Marg charge ₹50,000–₹5,00,000/year. Small clinics cannot afford enterprise software.
Too Fragmented	Pharmacy uses one system, lab uses another, doctor uses paper. No unified patient record exists.
Not India-Ready	Most software is built for Western workflows. GST billing, Indian drug databases, OTP-based auth, and INR billing are afterthoughts.

1.3 Our Solution

BHarath Health Systems is a **unified, cloud-based platform** that connects every department of a clinic or hospital into one system — accessible from any device, anywhere, at a fraction of the cost of legacy software.

■ Unified One platform for reception, doctor, pharmacy, lab, imaging, ward, and administration. Single patient record flows through all.	■ Affordable Subscription-based pricing designed for Indian clinics. Free tier for single-doctor clinics. No hardware needed.
■ ■ India-First Built with GST billing, Indian drug database (5,000+ drugs), OTP authentication, INR currency, and ICD-10 coding.	■ ■ Cloud-Native No installation, no server maintenance, no IT team needed. Works on any browser, any device.
■ Secure JWT authentication, role-based access control, PIN-gated clinical actions, audit logs for every change.	■ Scalable Multi-tenant architecture supports unlimited clinics on one backend. Built to scale to 1M+ users.

1.4 Vision Statement

"To become the default healthcare operating system for every clinic and hospital in India — making world-class clinical software accessible to a doctor in a village as easily as to a hospital chain in Mumbai."

1.5 Target Market

Segment	Size	Our Offer
Single-doctor clinics	~6,00,000 in India	Free/Basic plan — OPD, prescriptions, billing
Multi-doctor clinics	~1,50,000 in India	Standard plan — full OPD + pharmacy + lab
Small hospitals (10–50 beds)	~50,000 in India	Professional plan — inpatient + all modules
Mid-size hospitals (50–200 beds)	~15,000 in India	Enterprise plan — full suite + analytics

1.6 Revenue Model

Plan	Price/Month	Included
Free	■0	Up to 2 doctors, basic OPD + billing, 1 branch
Standard	■2,999	Up to 5 doctors, pharmacy + lab, 2 branches
Professional	■7,999	Up to 15 doctors, inpatient/ward, 5 branches
Enterprise	■19,999+	Unlimited doctors, all modules, unlimited branches

■ All plans include: GST-compliant billing, Indian drug database, OTP auth, unlimited patients, and free SSL/hosting. No hidden charges.

Chapter 2 — System Architecture Overview

2.1 High-Level Architecture

BHarath Health Systems uses a **monolithic-backend / multi-frontend architecture**. A single FastAPI backend serves all 9 portals via a unified REST API. Each portal is an independent React SPA that communicates exclusively through this API. The database is a single PostgreSQL instance with row-level multi-tenancy via `clinic_id`.

USERS / BROWSERS
Public visitors · Patients · Receptionists · Managers Doctors · Nurses · Pharmacists · Lab staff · Radiologists · Admins Any device — desktop, tablet, mobile
FRONTEND LAYER — 9 React (Vite) SPAs on Vercel / Cloudflare Pages
public patient staff provider carechart pharmacy lab imaging admin Each SPA: React 18 + React Router + Axios + Tailwind CSS Communicates via HTTPS to backend API only — no direct DB access
BACKEND LAYER — Single FastAPI App on Render
All routes under <code>/api/v1</code> · JWT authentication · Role-based guards ~140 inpatient routes · pharmacy · lab · imaging · billing · auth Rate limiting (slowapi) · CORS middleware · Security headers Python 3.11 · FastAPI 0.111 · SQLAlchemy ORM · Gunicorn + Uvicorn
DATABASE LAYER — PostgreSQL on Supabase
70+ tables · Multi-tenant via <code>clinic_id</code> on every row No Alembic — schema managed via <code>start.sh</code> (CREATE TABLE IF NOT EXISTS) NullPool connection — Supervisor manages pooling (45 usable connections) Auto-seeded: 5,000+ drugs · ICD-10 terms · Lab tests · Imaging catalog
EXTERNAL SERVICES
Fast2SMS / Twilio — OTP delivery · Resend — email notifications Cloudinary — image/document storage · Daily.co — telehealth video GitHub Actions — CI/CD · Sentry — error monitoring (planned)

2.2 Request Flow Example

Example: A receptionist books an appointment for a walk-in patient.

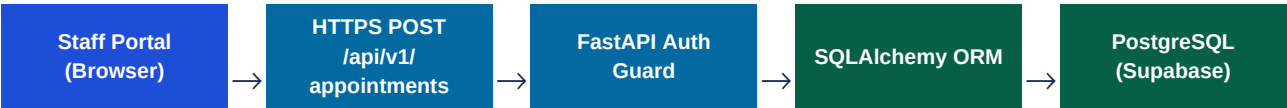


Figure 2.1 — Request flow from browser to database

2.3 Multi-Tenancy Design

Every table that contains clinic-specific data has a `clinic_id` foreign key. The backend resolves `clinic_id` from the authenticated staff JWT on every request. This means a pharmacist at Clinic A can never read or write data belonging to Clinic B — isolation is enforced at the SQL query level, not the application level.

```
Example: SELECT * FROM appointments WHERE clinic_id = :clinic_id AND id = :id
```

■ `org_type` field on the `Clinic` model controls feature access: "clinic" gets OPD features, "hospital" additionally gets inpatient/ward modules.

2.4 The 9 Portals — Overview

#	Portal	Directory	Users	URL
1	Public	frontend/public/	Visitors, patients	bharathhealthsystems.com
2	Patient	frontend/patient/	Registered patients	my.bharathhealthsystems.com
3	Staff	frontend/staff/	Receptionists + Managers	staff.bharathhealthsystems.com
4	Provider	frontend/provider/	Doctors	provider.bharathhealthsystems.com
5	CareChart	frontend/carechart/	Nurses, ward staff	carechart.bharathhealthsystems.com
6	Pharmacy	frontend/pharmacy/	Pharmacists	pharmacy.bharathhealthsystems.com
7	Lab	frontend/lab/	Lab technicians	lab.bharathhealthsystems.com
8	Imaging	frontend/imaging/	Radiologists	imaging.bharathhealthsystems.com
9	Admin	frontend/admin/	Platform superadmins	(internal)

Chapter 3 — Portal 1: Public Portal

PORTAL 1

Public Portal

bharathhealthsystems.com

3.1 Purpose

The Public Portal is the **front door of BHarath Health Systems**. It serves two distinct audiences: (1) **Patients and the public** who want to find doctors, book appointments, or access health information — without creating an account first. (2) **Clinics and hospitals** who want to register their facility on the platform and start using the system. No login is required for browsing; OTP-based verification is used for booking.

3.2 Key Features

■ Doctor Discovery

Search doctors by name, specialization, clinic, or location. View doctor profiles, qualifications, consultation fees, and available time slots.

■ Online Appointment Booking

Book appointments without creating an account. OTP verification via mobile ensures authenticity. Patients receive confirmation.

■ Clinic Discovery

Search for clinics and hospitals by location, specialty, or name. View clinic details, branches, working hours, and available doctors.

■ Clinic Self-Registration

New clinics register themselves on the platform. Form captures clinic name, type (clinic/hospital), specialties, branch details, and admin contact. Submitted for platform admin approval.

■ Pre-Visit Forms

Patients fill health questionnaires before arriving — chief complaint, existing conditions, current medications. Saves time at reception.

■ Public Health Information

Marketing pages explaining platform features, pricing plans, and benefits for clinics and hospitals.

3.3 Pages & Routes

Route	Page	Description
/	Landing Page	Hero, features, pricing, CTA to book or register
/find-doctor	Doctor Search	Search + filter doctors, view profiles
/book/:doctorId	Booking Flow	Select slot → OTP verify → confirm booking
/clinic/:clinicId	Clinic Profile	Clinic info, branches, doctors, hours
/register-clinic	Clinic Registration	Multi-step form for new clinic signup
/pre-visit/:token	Pre-Visit Form	Patient health questionnaire before visit
/plans	Pricing	Subscription plans and feature comparison

3.4 API Endpoints Used

The Public Portal uses **unauthenticated endpoints only** — no JWT required. All routes are under **/api/v1/public/** and **/api/v1/otp/**.

Method	Endpoint	Purpose
GET	/api/v1/public/doctors	Search/list doctors with filters
GET	/api/v1/public/doctors/:id/slots	Get available appointment slots
POST	/api/v1/public/book	Book an appointment (OTP-verified)
GET	/api/v1/public/clinics	Search clinics/hospitals
GET	/api/v1/public/patient-lookup	Check if patient exists by mobile
POST	/api/v1/otp/send	Send OTP to mobile number
POST	/api/v1/otp/verify	Verify OTP and return verified token
POST	/api/v1/public/register-clinic	Submit new clinic registration
GET	/api/v1/public/patient-profile	Get patient profile by verified token

■ OTP routes are mounted at /api/v1/otp/ — NOT at /otp/. This is a common mistake. The /api/v1 prefix always applies.

3.5 Booking Flow — Step by Step

Example: A patient named Ravi books an appointment with Dr. Sharma.

Step	Action	Technical Detail
1	Patient visits /find-doctor, searches "Cardiologist"	GET /public/doctors?specialization=Cardiologist
2	Selects Dr. Sharma, clicks "Book Appointment"	GET /public/doctors/42/slots?booking_date=2026-06-22

3	Selects 10:30 AM slot, enters mobile 9876543210	POST /otp/send { mobile: "9876543210" }
4	Receives OTP 4521 on mobile, enters it	POST /otp/verify { mobile: "9876543210", otp: "4521" } → returns verified_token
5	Booking confirmed	POST /public/book { doctor_id: 42, slot: "10:30", verified_token: "..." }
6	Patient gets SMS confirmation	Backend sends confirmation via Fast2SMS

3.6 Clinic Registration Flow

When a new clinic wants to join the platform:

- 1. Clinic admin visits /register-clinic and fills the registration form
- 2. Form captures: clinic name, type (clinic/hospital), specialties, branch address, admin name, email, mobile
- 3. POST /public/register-clinic → creates a pending Clinic record in the database
- 4. Platform Admin receives notification and reviews the registration
- 5. Admin approves → clinic status changes to "active" → admin credentials emailed
- 6. Clinic admin logs into Staff Portal to set up doctors, schedules, and settings

3.7 Database Tables Used

Table	Purpose	Key Fields
clinics	Clinic/hospital records	id, name, org_type, status, clinic_id
branches	Clinic branch locations	id, clinic_id, address, city, phone
doctor_profiles	Doctor information	id, staff_id, specialization, fee, bio
doctor_schedules	Availability slots	id, doctor_id, day_of_week, start_time
appointments	Booked appointments	id, clinic_id, patient_id, doctor_id, status
patients	Patient records	id, clinic_id, name, mobile, bh_id
online_bookings	Public bookings	id, verified_token, appointment_id, status

3.8 Technical Stack

Frontend directory: **frontend/public/**

Component	Technology	Notes
Framework	React 18 + Vite	SPA, no SSR
Routing	React Router v6	Client-side routing
HTTP Client	Axios	Configured in src/api/client.js
Styling	Tailwind CSS	Utility-first, custom brand colors

Auth	None (public)	OTP verified_token for booking only
State	React useState/useEffect	No global state library needed
Deployment	Vercel	Auto-deploy from main branch
Env Var	VITE_API_URL	Points to https://bharatcliniq-api.onrender.com

■ *The public portal has NO JWT authentication. Any endpoint it calls must be accessible without an Authorization header. Never add protected endpoints here.*

3.9 Key Design Decisions

◆ OTP instead of password

Patients in India are comfortable with OTP-based auth (used by banks, UPI, etc.). No password to forget, no account to create.

◆ No login required for browsing

Reduces friction for new patients. They can explore doctors and clinics before committing to any interaction.

◆ Verified token pattern

After OTP verification, the server returns a short-lived verified_token. This token is passed with the booking request to prove the patient is who they claim to be — without creating a full session.

◆ Unauthenticated API only

The public portal never handles staff JWTs or patient session tokens. This limits the attack surface significantly.